



Sampling Rate and Bit Depth

Audio Features and Convolutional Neural Networks



Video: Audio Feature Extraction
10 min



Video: Introduction to Convolutional Neural Networks
11 min



Video: Modifying the Neural Network
10 min



Reading: MFCCs and CNNs
5 min



Practice Quiz: MFCCs and CNNs
5 questions

Deployment to Embedded Systems



Video: Deploy Keyword Spotting System
6 min



Video: Implementation Strategies
10 min



Video: Sensor Fusion
3 min



Reading: Implementation Strategies and Sensor Fusion
5 min



Practice Quiz: Implementation Strategies
5 questions

Project and Review



Reading: Project - Sound Classification
2h



Quiz: Audio Classification
10 questions



MFCCs and CNNs

While optional, we encourage you to take a look at the following articles to learn more about the topics covered in this lesson:

Video that explains the Fourier Transform:

<https://www.youtube.com/watch?v=spUNpyF58BY>

[The Nyquist-Shannon Theorem: Understanding Sampled Systems](#)

[Sampling Theorem and Frequency Spectrum Aliasing](#)

[Mel Frequency Cepstral Coefficient \(MFCC\) tutorial](#)

[Speech Processing for Machine Learning: Filter banks, Mel-Frequency Cepstral Coefficients \(MFCCs\) and What's In-Between](#)

[A Comprehensive Guide to Convolutional Neural Networks — the ELI5 way.](#)

[Convolutional Neural Networks \(CNN\): Step 1\(b\) - ReLU Layer](#)

[1D Convolutional Neural Networks and Applications – A Survey](#)

[A Gentle Introduction to Dropout for Regularizing Deep Neural Networks](#)

[How to Configure the Learning Rate When Training Deep Learning Neural Networks](#)

✓ Complete

Go to next item

